535/1
PHYSICS
Paper 1
(Theory)
Jul./Aug. 2026
2½ hours

WAKISO-KAMPALA TEACHERS' ASSOCIATION (WAKATA)

WAKATA MOCK EXAMINATIONS 2026

Uganda Certificate of Education

PHYSICS

WAKATA 2026

Paper 1
(Theory)

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

This examination paper consists of **two** *Sections;* **A** *and* **B**.

It has **seven** *examination items.*

Section **A** *has* **three compulsory** *items.*

Section **B** *has* **two** *Parts;* **I** *and* **II**. *Respond to* only **one** *item from each part.*

Respond to **five** *items in all.*

Any additional item(s) responded to will **not** *be scored.*

All responses must be written in the answer booklet(s) provided.

Responses to each item should start on a fresh page.

Indicate the item number you are responding to, clearly at the top of the page.

Graph paper is provided.

Mathematical tables and silent non-programmable calculators may be used.

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SECTION A

Respond to **all** *the items in this section*

Item 1.

An old man drove to the hospital to see an optician as he can no longer read his books without eye glasses that broke down recently. He was accompanied by a boy who needed a tooth extraction. In the parking lot, the boy was puzzled to see that the words on an emergency vehicle were written backwards. As they walked through the hospital, they met a woman who asked the old man for some guidance. At the dental clinic, the dentist asked the nurse to select an optical instrument that would best extract the tooth.

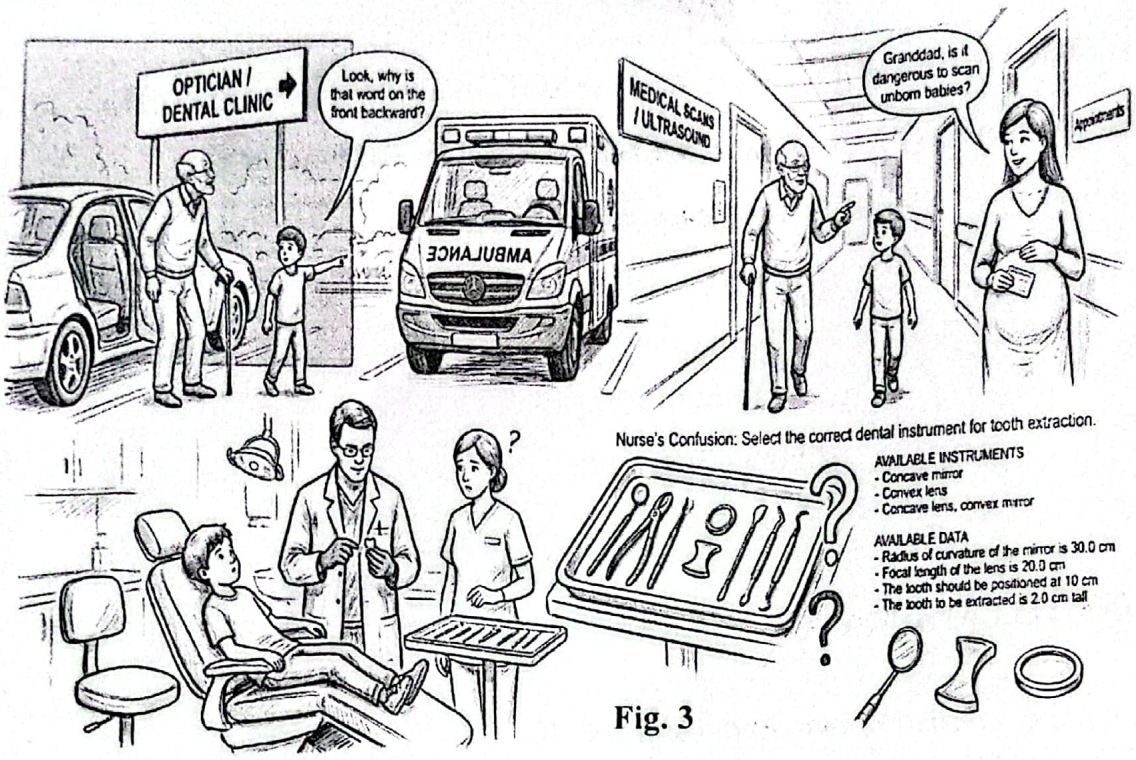


Fig. 3

Task: Study the scene in the picture and help;

(a) (i) choose the suitable instrument for the tooth extraction. (*Support your choice using graphical analysis to determine the magnification of the tooth*)

(ii) explain why the words on the vehicle appeared that way.

(b) (i) identify the lens that would solve the eye defect.

(ii) identify applications of the other optical instruments that were not appropriate for use.

(iii) the old man guide the woman by explaining how the process works.

Item 2.

At a certain hospital, a patient was treated for a thyroid disorder. The patient was carefully monitored for safety, and the medical team explained that the radiation decreases over time. To ensure the patient didn't expose his relatives to harmful radiation, they measured the radiation activity over time *(see table 1)*. The substance is safe when the activity drops below the background radiation. Despite the data presented to the relatives, they could not understand anything and wondered if radiation are only important for use in the hospital.

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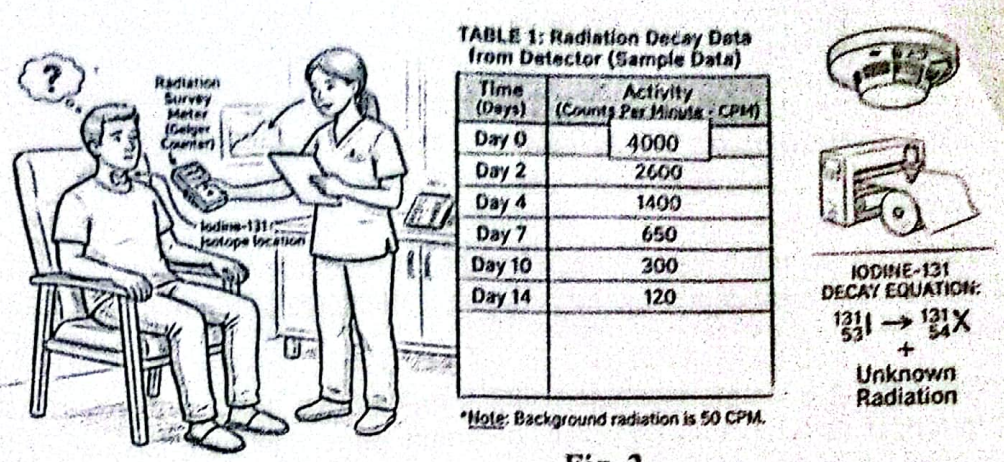


TABLE 1: Radiation Decay Data from Detector (Sample Data)

Time (Days)	Activity (Counts Per Minute - CPM)
Day 0	4000
Day 2	2600
Day 4	1400
Day 7	650
Day 10	300
Day 14	120

*Note: Background radiation is 50 CPM.

Fig. 2

Task: Help the relatives;

(a) Understand how isotope used for treating the patient works.

(b) Know when they should come back to check the patient in the isolation room

(c) Briefly explain the applications of the radiation pictured in the scene.

Item 3.

During a video call between a girl from Uganda and her friend who was on a summer trip in Norway, an incredible phenomenon happened. As they were talking, the friend pointed her camera out of the window. The digital clock on the friend's wall in Norway read 12:30 a.m., yet outside, the sun was still shining brightly in the sky. The girl in Uganda looked out her own window, where it was completely dark, and asked in surprise: "How is the sun still shining over there in the middle of the night?"

As they were chatting, the friend stood up and walked close to a corner of the room. Immediately, a light in the background turned on by itself. The girl was surprised and asked, "How did that light turn on without you touching a switch?"
The friend explained that it was an automatic lighting system. She then showed her a simple circuit diagram (*see figure* 3.)

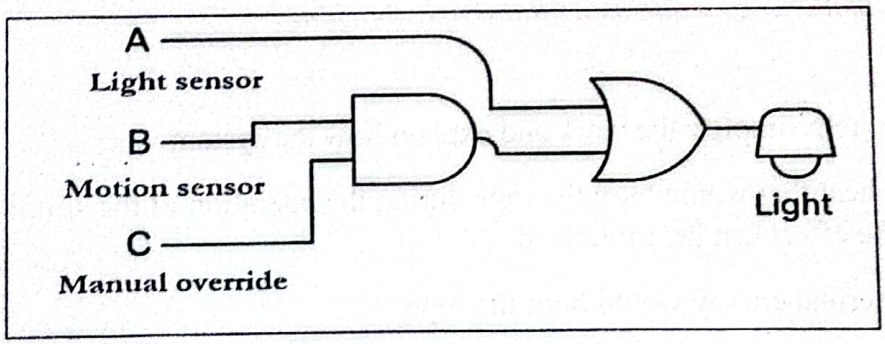


Fig. 3

The girl was concerned about:

(a) How the system works

(b) Cause of the solar cycle, and why Norway experiences different seasons

(c) Application of satellites in their communication.

Task: As a learner of Physics, help the girl understand her concerns.

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SECTION B

PART I

Respond to **one** *item from this part*

Item 4.

During a dry season, a man is manually excavating a borehole to provide clean water for his community. He reached a significant depth, but he is facing a challenge to remove the heavy clay soil to the surface.

His nephew suggested making a simple machine using a rope which had a safety tag that it will suffer thermal failure if its temperature exceeds a certain value. The man is worried that pulling the heavy bucket up will generate enough heat to burn the rope. The nephew asked about the properties of the building materials that were chosen to construct the borehole.

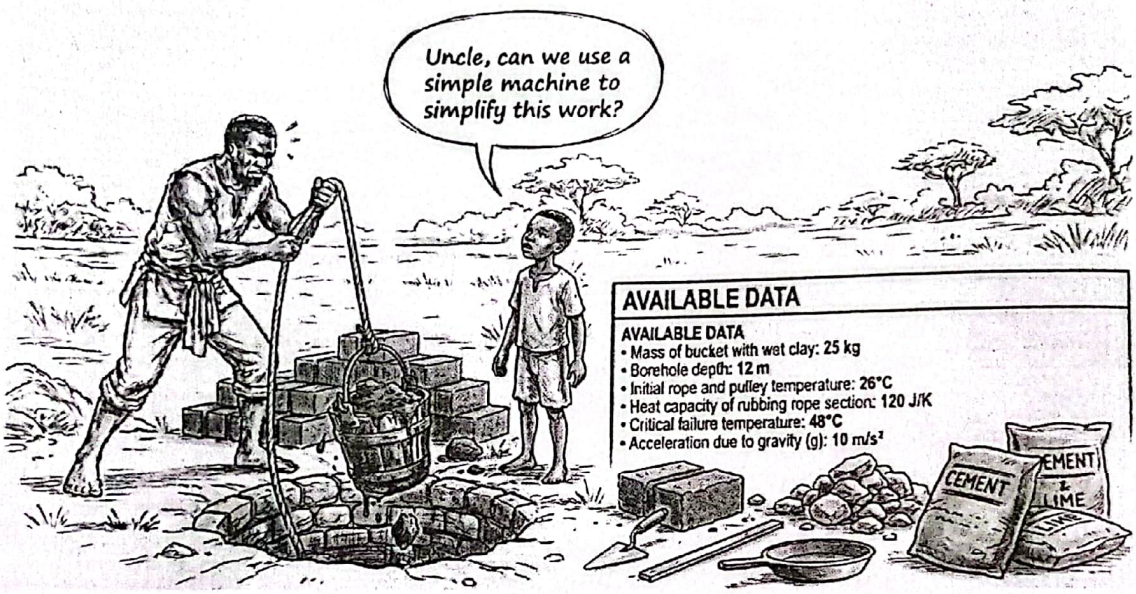


Fig. 4

Hint: *Assume all mechanical work done against gravity in lifting the heavy bucket of soil transforms entirely into heat energy while using the simple machine.*

Task: Help the man;

(a) Design a system that can help simplify the work and explain how the system works.

(b) Explain the origin of the heat that would burn the rope during the operation of the simple machine and state how the effect can be minimized.

(c) Determine whether the thermal energy would burn the rope.

(d) Briefly explain the mechanical properties of the building materials available on site.

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Item 5.

Early in a very cold morning, a rider was hired to deliver a package to the a nearby lakeside town. Despite the coldness, the rider forgot his leather riding jacket and safety helmet at home. He rode straight to his destination, and his journey data was recorded by a mobile tracking app as follows:

- *He accelerated from rest to 50kmh^{-1} in 15 minutes and maintained the speed for 1 hour*
- *Then accelerated for 15 minutes until he reached 80 kmh^{-1}*
- *He then decelerated to 20 kmh^{-1} over 15 minutes*
- *And finally decelerated to rest in 15 minutes*

Upon arriving, the rider was shivering and had developed a fever from exposure to the cold wind. He demanded his agreed payment of 60,000 UGX, which was the rate for a full 100 km journey. However, the shop owner looked at the tracking app and refused to pay the full amount. Furthermore, the shop owner warned the rider for risking his health by not wearing his safety gear.

Task: Help the rider;

(a) Understand;

(i) why riding without a jacket caused him to develop a fever, and how a thick jacket prevents this.

(ii) how the inner foam lining of a helmet protects a rider's head during incase of accident.

(b) Know if he deserves the whole pay for the journey.

(c) Explain any application of convection as a means of heat transfer in real life.

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PART II

Respond to **one** *item from this part*

Item 6.

During a trip to a car showroom, learners were introduced to a hybrid car. The guide explained that this vehicle uses both a petrol engine for high-speed driving and an electric motor, powered by a battery, for low-speed driving. The switch between fuel and electricity is done automatically. When the car slows down, the relay disconnects the fuel system and activates the electric motor.

The learners inquired about the working of the automatic switch, whether the current that flows from the battery can turn the switch on and explanation of the working of the motor.

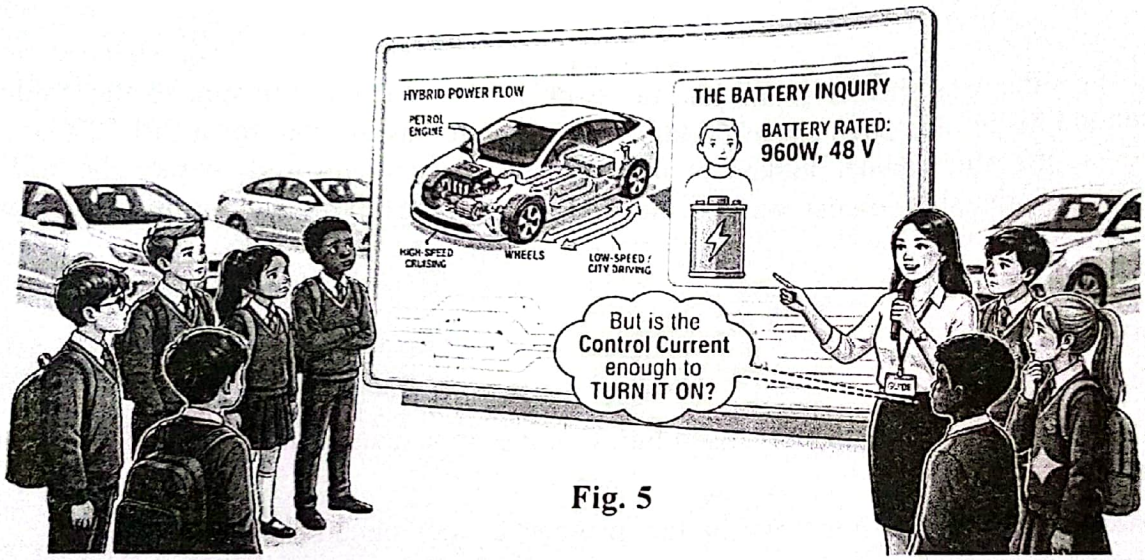


Fig. 5

The guide mentioned that the hybrid car battery should deliver a minimum current of 17A to start the electric motor.

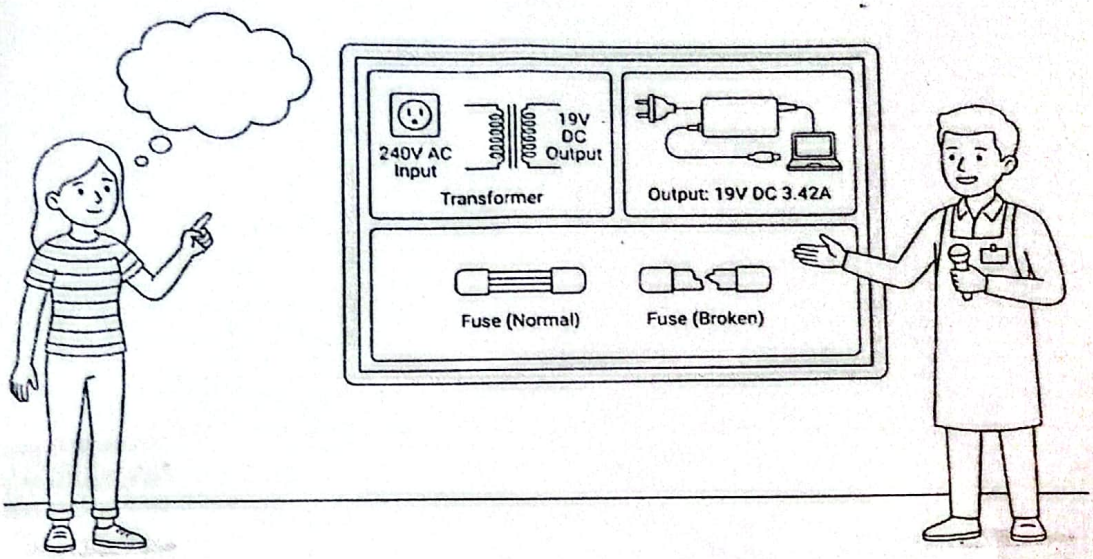
Task: Help the learners to understand their concern

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Item 7.

A student's laptop suddenly stopped charging, prompting her to visit a local technician. After a quick inspection, the technician advised her that the laptop itself was fine, but she needed to purchase a new power adapter.

At the electronics shop, she was handed a new adapter. Before buying it, she carefully examined the technical specifications printed on the plastic casing:



The student became curious and concerned about the following statements from the shopkeeper;

- *Laptops charge on DC, but electricity from the wall socket is AC.*
- *The adapter is specified with 19 V output, but domestic electricity supply is 240 V*
- *The adapter fuse protects it from large flow of current.*

Task: Help the student understand the statements from the shopkeeper.

END

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